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Superseding
(see 6.5)

MILITARY SPECIFICATION

COUPLER, OPTICAL

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of an optical coupler, used on the M981 Fire Support Team Vehicle (FISTV).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1240

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SPECIFICATIONS
FEDERAL

BB-N-411	- Nitrogen, Technical.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coating on Aluminum and Aluminum Alloys.
MIL-O-13830	- Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of.
MIL-P-23377	- Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-11268	- Parts, Materials and Processes used in Electronic Equipment.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-965	- Parts Control Program.
MIL-STD-1523	- Age Controls of Age-Sensitive Elastomeric Material.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

11507293	- Rangefinder - Target Designator, Laser MX-9759/TVQ-2 (G/VLLD).
12310400	- Optical Coupler Assembly.
12343606	- ITA Assembly.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish optical couplers which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent optical couplers delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. The materials, processes and finishes used for the manufacture of the optical coupler shall be of high quality, suitable for the intended purpose under the requirements imposed herein (see 4.6.1).

3.2.1 Metals. All metals utilized shall be of corrosion resistant type or suitably protected to resist corrosion during the normal service life under the environmental conditions specified. The use of dissimilar metals shall be in accordance with MIL-STD-889.

3.2.2 Nonmetals. All nonmetals shall be suitable for their intended purpose during the normal life and under the environmental conditions specified and shall be in accordance with MIL-STD-454 requirements 3 (Flammable) and 4 (Fungus-Inert). Use of elastomerics shall be governed by the age controls of MIL-STD-1523.

3.2.3 Paint. All exterior painted surfaces shall be painted with chemical agent resistant aliphatic polyurethane coating conforming to MIL-C-46168, Color No. 383 Green. All aluminum surfaces shall be cleaned and treated per MIL-C-5541 and primed per MIL-P-23377, Type I.

3.3 Design and construction. The optical coupler shall provide the means for relaying the image ray bundle emerging from the exit pupil of the Ground/Vehicular Laser Locator and Designator (G/VLLD, 11507293) into the Image Transfer Assembly (ITA, 12343606) so that an operator can look into the ITA eyepiece and see through the G/VLLD. The input to the coupler is a collimated image ray bundle from the G/VLLD eyepiece. The output of the coupler is an image located at the paraxial image plane within the ITA. The optical coupler shall permit easy dismounting of the G/VLLD from the vehicle without removing the coupler from the ITA.

3.3.1 Configuration. The optical coupler shall be fabricated and assembled in accordance with Drawing 12310400, and parts lists and other documents listed herein (see 4.6.2).

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3.3.2 Protective window. The optical coupler objective lens shall be covered by a protective window with a protection level equivalent to BK 7 glass or better (see 4.6.1).

3.3.3 Parts. The selection of parts shall be based on the requirements of MIL-STD-965. Parts selection shall be in accordance with the following order of precedence (see 4.6.1):

a. Standard parts as defined by MIL-P-11268.
b. The TACOM Military Component Supply Directory.
c. Ordnance.
d. Federal specifications and standards.
e. Industry standards.

3.3.4 Weight. The weight of the optical coupler shall be no heavier than 9.75 pounds (lb) [4.42 kilograms (kg)] (see 4.6.3).

3.3.5 Leak resistance. The helium leak rate for the optical coupler assembly, when mounted to the ITA, or a fixture simulating the ITA, shall be no greater than 0.000025 atmosphere-cubic centimeter (cc) per second when tested with an internal atmosphere which is 10 $\pm$ 1 percent (%) helium with an ITA (or fixture) internal pressure of one atmosphere and an external pressure which is no greater than 1/1500 atmosphere [67.55 Pascal (Pa)] (see 4.6.4).

3.3.6 Purge and charge. The optical coupler shall be purged of all the air and the moisture, and charged to an internal pressure of 0 $\pm$ 0.2 pounds per square inch gage (psig) [0 $\pm$ 1.38 kilopascals (kPa)] with dry nitrogen conforming to BB-N-411, Type I, Class 1, Grade A or B (see 4.6.1).

3.4 Performance. The optical coupler shall output the image quality required in 3.4.1.1 through 3.4.7 as a component, or as part of the integrated system, as specified.

3.4.1 Resolution. Resolution specified in 3.4.1.1 and 3.4.1.2 shall be for a high contrast target which shall be the USAF 1951 test chart or the NBS 1010A microcopy test chart.

3.4.1.1 Optical coupler resolution. The optical coupler shall clearly resolve an angle of 104 arc seconds or less, on axis (see 4.6.5.1).

3.4.1.2 Integrated system resolution. When the optical coupler is assembled between the G/VLLD and the ITA, the system shall clearly resolve an angle of 8 arc seconds or less, on axis, with the observer's eye anywhere within the exit pupil of the ITA eyepiece. The integrated system shall also clearly resolve the G/VLLD LED data numbers which are approximately in the plane of the G/VLLD reticle (see 4.6.5.2).

3.4.2 Astigmatism. The optical coupler axial astigmatism shall not be greater than 0.2 diopter (see 4.6.6).

3.4.3 Magnification. The optical coupler shall provide a magnification of 1 $\pm$ 7% from the optical coupler input to the ITA output (see 4.6.7).

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3.4.4 Entrance pupil. The optical coupler shall provide an entrance pupil that shall not reduce the integrated system exit pupil to less than 0.22 inch (in) [5.5 millimeters (mm)] when the integrated system is properly pupil matched (see 4.6.8).

3.4.5 Field of view. The integrated system, when incorporating the optical coupler, shall provide a real field of view that is not less than 5 degrees (.087 radians) in any pupil matched condition (see 4.6.9).

3.4.6 Lateral color. The optical coupler shall provide that the lateral color of the system shall not be greater than 3 minutes of an arc out of the system for wavelengths between .0000192 in [486 nanometer (nm)] and .0000258 in (656 nm) (see 4.6.10).

3.4.7 Visual anomalies. The optical coupler shall contribute no visually perceptible defects to the integrated system such as flare ghost images or effects outside the ITA exit pupil (see 4.6.11).

3.4.8 Environmental conditions. The optical coupler shall function as specified when exposed to the environmental conditions specified herein. The optical coupler shall meet the performance requirements of 3.4.1 through 3.4.7 during exposure to those environments designated as operating. The optical coupler shall not be damaged and shall meet requirements after exposure to those environments designated as nonoperating or storage.

3.4.8.1 Temperature. The optical coupler shall operate in ambient air temperature of +145 degrees Fahrenheit (°F) [63 degrees Celsius (°C)] to -25°F (-32°C). The optical coupler shall be capable of being stored at temperatures of +160°F (71°C) to -40°F (-40°C) (see 4.6.12).

3.4.8.2 Temperature shock. The optical coupler shall not show any operational degradation after being exposed to temperature shock in accordance with MIL-STD-810, Method 503.2, Procedure 1, with temperature extremes of -40°F (-40°C) to +160°F (71°C) (see 4.6.13).

3.4.8.3 Humidity. The optical coupler shall operate in humidity conditions of 95% relative humidity (rh) at temperatures from -25°F (-32°C) to 145°F (63°C); and at humidity conditions of 5% rh at temperatures from +70°F (21°C) to +145°F (63°C) (see 4.6.14).

3.4.8.4 Ambient pressure. The optical coupler shall operate at ambient air pressure of 14.7 to 10.1 pounds per square inch absolute (psia) (100 to 69.6 kPa) which corresponds to altitudes of 0 to 10 000 feet [0 to 3.2 kilometers (km)]. Nonoperating, the optical coupler shall withstand exposure to air pressures from 14.7 to 2.47 psia (100 to 17 kPa) which corresponds to altitudes of 0 to 40 000 feet (0 to 12.2 km) (see 4.6.15).

3.4.8.5 Rain (nonoperating). The optical coupler shall operate without degradation of performance after exposure to a 12 hour, 9.50 in [24.1 centimeters (cm)] rainfall which includes intermittent winds of 35 knots [65 kilometers per hour (km per h)] and the following intensities (see 4.6.16):

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Duration	Rainfall rate
1 minute	27 inches per hour (in per hr) [68.6 centimeter per hour (cm per hr)]
5 minutes	12 in per hr (30.5 cm per hr)
10 minutes	9 in per hr (23 cm per hr)
1 hour	5.5 in per hr (14 cm per hr)

3.4.8.6 Sand and dust. The optical coupler shall be operational while exposed to windblown particles 0.000394 to 0.0394 in (0.01 to 1 mm) in diameter with predominant diameters between 0.00591 and 0.01181 in (0.15 and 0.30 mm) with winds greater than 30 knots (56 km per h) (see 4.6.17).

3.4.8.7 Fungus (nonoperating). The optical coupler shall be operational after being exposed to the fungi specified in MIL-STD-810, Method 508.3, for 90 days at a relative humidity of 95% at 84°F (29°C) (see 4.6.18).

3.4.8.8 Vibration (nonoperating). The optical coupler shall not be damaged nor performance degraded after exposure to vibration levels of MIL-STD-810, Method 514.3, Category 8, Procedure I, as specified in figure 3, with sinusoidal cycling time of 3 hours per axis in accordance with the following schedule. The sweep time shall be 15 minutes from 5 to 500 to 5 Hz (see 4.6.19).

Frequency	Vibration level
5 to 5.5 Hertz (Hz)	1 inch double amplitude
5.5 to 30 Hz	1.5 gravity (g)
30 to 50 Hz	0.033 inch double amplitude
50 to 500 Hz	4.2 g

3.4.8.9 Shock (nonoperating). The optical coupler shall not be damaged nor performance degraded after exposure to shock levels in accordance with MIL-STD-810, Method 516.3, Procedure I. Figure 516.3-4 with an 11 millisecond (ms) sawtooth pulse of 40 g (peak) shock shall be applied (see 4.6.20).

3.5 Identification and marking. The optical coupler assembly shall be marked per MIL-STD-130 and MIL-P-514, and as indicated on Drawing 12310400 and lower tier drawings (see 4.6.2).

3.6 Workmanship. Workmanship shall be of a quality to assure that the optical coupler and its components are free of defects that compromise, limit or reduce capability in the performance of its intended use. No workmanship condition shall exist which may create a safety hazard to operating or maintenance personnel. Particular attention shall be paid to the marking of parts, plating and freedom of parts from burrs and sharp edges. Visible finger prints, smudges or internal moisture shall be cause for rejection. Scratches, digs, bubbles, inclusions, and cement defects exceeding criteria stated in sections 3.5 and 3.6 of MIL-O-13830 shall be cause for rejection (see 4.6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or this specification (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or this specification, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Qualified products. In the event that particular specifications referenced in this document have a Qualified Products List (QPL), the supplier shall utilize items only from vendors specified in the applicable QPL for the manufacture of production items. The supplier shall document the acquisition of all QPL items with listings of suppliers, quantity and date of QPL. Such QPL documentation shall be made available to the Government upon request.

4.1.2 Quality assurance requirements (QAR)/supplementary quality assurance provisions (SQAP). In the event that ordnance or MS drawings referenced in this document or lower tier drawings applicable to the referenced drawings have QARs or SQAPs applicable, the supplier shall inspect the parts/components in accordance with the QAR and SQAP requirements. In addition, General Quality Assurance Requirements (STA Form 4452) also apply in conjunction to the QAR/SQAP requirements. The supplier shall submit QARs/SQAPs to subcontractors with purchase orders for items that have QAR/SQAP requirements.

4.1.3 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.1).
    2. Acceptance tests (see 4.5.2).
    3. Control tests (see 4.5.2.2).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 ± 10°C (73 ± 18°F)
b. Barometric pressure 725^{+50}_{-75} mm Hg (28^{+2}_{-3} inches Hg)
c. Relative humidity 50 ± 30 percent

4.4 First article tests and inspection.

4.4.1 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply optical coupler assemblies that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.2 In-process inspection. During fabrication of the first production optical coupler, in-process inspections will be conducted by Government representatives to evaluate conformance of materials and workmanship to requirements of specified documents. The coupler and all components shall be available for inspection. These inspections shall be made at the contractor or subcontractor's facilities prior to application of primer and paint. Processing and welding procedures, quality system, inspection records, calibration procedures, radiographic procedures and welder certifications will be reviewed and evaluated during the in-process inspections.

4.4.3 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two optical coupler assemblies. Preproduction inspection shall consist of inspection as specified in table I.

4.4.4 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two from the first ten optical coupler assemblies produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table I.

4.4.5 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prima facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government. The Government shall not final accept products until first article testing is completed to the satisfaction of the Government.

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TABLE I. Classification of inspections and tests.

Title	Requirement	Inspection	First article	Quality conformance	
				Acceptance (100%)	Control
Materials and construction	3.2 thru 3.2.3, 3.3.2, 3.3.3, 3.3.6 and 5.1	4.6.1	X	X	
Weight	3.3.4	4.6.3	X		X
Leak resistance	3.3.5	4.6.4	X	X	
Defects (see table II)		4.6.2	X	X	
Resolution					
Optical coupler resolution	3.4.1.1	4.6.5.1	X	X	
Integrated system resolution	3.4.1.2	4.6.5.2	X	X	
Astigmatism	3.4.2	4.6.6	X	X	
Magnification	3.4.3	4.6.7	X		X
Entrance pupil	3.4.4	4.6.8	X		X
Field of view	3.4.5	4.6.9	X		X
Lateral color	3.4.6	4.6.10	X		X
Visual anomalies	3.4.7	4.6.11	X		X
Temperature	3.4.9.1	4.6.12	X		
Temperature shock	3.4.9.2	4.6.13	X		
Humidity	3.4.9.3	4.6.14	X		
Ambient pressure	3.4.9.4	4.6.15	X		
Rain	3.4.9.5	4.6.16	X		
Sand and dust	3.4.9.6	4.6.17	X		
Fungus	3.4.9.7	4.6.18	X		
Vibration	3.4.9.8	4.6.19	X		
Shock	3.4.9.9	4.6.20	X		

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the optical coupler assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

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4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective (see 6.4).

Classification	AQL
Major	1.5
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.5% Defective	
101	Housing; structural and welding defects.	Visual and functional
102	Covers, seals and windows; fit improper, malfunction, leaks, damage, non-conformance.	Visual and functional
103	Lenses, prisms, mirrors, reticles and windows; damage, malfunction.	Visual and
Minor	AQL 2.5% Defective	
201	Protective coatings; application, materials and coverage improper.	Visual
202	Records, forms or publications; missing, improperly processed or secured.	Visual
203	Fasteners; improper assembly, torque.	Visual and functional
204	Covers, seals and windows; improper assembly or installation.	Visual and functional
205	Lenses, prisms, mirrors, reticles and windows; improper assembly, installation or clearance.	Visual and functional
206	Paint; application or color improper.	Visual
207	Decals, marking data and instruction plates; missing, improper location or dize, defective.	Visual

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each optical coupler assembly shall be subjected to the tests specified in table I.

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4.5.3.2 Control tests. Control tests shall be conducted on one optical coupler assembly from each lot of 250 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The optical coupler assemblies shall be subjected to the tests specified in table I.

4.5.3.3 Failure. Failure of any optical coupler assembly to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. To determine conformance to 3.2 through 3.2.3, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.6.2 Defects. Conformance to 3.3 through 3.3.6 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment. Each optical coupler shall be examined visually to determine that dimensions, configurations, parts and assembly conform to the requirements of Drawing 12310400. There shall be no loose, damaged or missing assemblies, components or attaching hardware. All moving parts shall operate without sticking, binding or interference. Exposed surfaces shall not have dirt or grease build-up; chipped, scratched or peeling paint; rust; corrosion or incorrect finish. The coupler shall contain no chips, burrs, sharp edges, process residue or other contamination. All seals and gaskets shall be properly sealed. All lenses, prisms, mirrors, and windows shall be free of visible finger prints, smudges and internal moisture. Scratches, digs, bubbles, inclusions and cement defects shall not exceed criteria stated in sections 3.5 and 3.6 of MIL-O-13830. The AQL's depicted in table II are listed for sampling purposes in the event sampling of end items is authorized by the Government.

4.6.3 Weight. To determine conformance to 3.3.4, the optical coupler shall be weighed and shall be no heavier than 9.5 lbs (4.31 kg).

4.6.4 Leak resistance. To determine conformance to 3.3.5, the optical coupler shall be placed in a suitable chamber wherein the helium leak rate from the coupler shall be measured. The internal pressure of the coupler shall be 1 atmosphere (atm) which includes 10 $\pm$ 1% helium. The chamber shall be pumped to vacuum so the external pressure on the coupler shall be no greater than 1/1500 atm (67.55 Pa). The helium leak rate shall be measured and shall not be greater than 0.000025 atm-cubic centimeters per second.

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4.6.5 Resolution.

4.6.5.1 Optical coupler resolution. To determine conformance to 3.4.1.1, the optical coupler shall be mounted in a test fixture as shown in figure 1. Using the USAF 1951 test chart or the NBS 1010A microcopy test chart, the high contrast target shall be brought into sharp focus. The smallest resolved pattern shall be noted and the angle of resolution in the object plane shall be calculated. An optical magnifier (microscope) may be used to aid in the observation. The coupler shall clearly resolve an angle of 104 arc seconds or less on axis.

4.6.5.2 Integrated system resolution. To determine conformance to 3.4.1.2, the optical coupler, G/VLLD, ITA and collimator shall be mounted onto the alignment fixture as shown in figure 2. The high contrast test target shall be brought into sharp focus at the intersection of the G/VLLD reticle crosshairs, if a G/VLLD is used. An optical magnifier or dioptometer may be used to aid the observation at the ITA eyepiece. The resolved angle shall be 8 arc seconds or less, on axis, with the observer's eye anywhere within the exit pupil of the ITA eyepiece. The G/VLLD, if used, shall be electrically connected and the lens cover shall be closed to prevent firing of the laser. The G/VLLD LED data numbers shall appear as zeros and shall be clearly resolved, if a G/VLLD is used.

NOTE: Wherever the G/VLLD is referenced in a test in 4.6, an optical test fixture which is equivalent to the G/VLLD may be used in its place.

4.6.6 Astigmatism. To determine conformance to 3.4.2, the optical coupler shall be mounted as shown in figure 1. The image of the astigmatism target shall be brought into sharp focus. When the image is viewed through an optical magnifier (microscope), readings shall be made when both the vertical lines of the astigmatism target are in focus and when the horizontal lines of the astigmatism target are in focus. The astigmatism, in diopters, is the difference between the two readings. A difference of 0.01 in (0.254 mm) corresponds to 0.2 diopter. The astigmatism shall not be greater than 0.2 diopter.

4.6.7 Magnification. To determine conformance to 3.4.3, install the optical coupler on the ITA. A telescope coupled to the ITA eyepiece shall be used to bring the target image into sharp focus at the center of the field. The angular size of the target image shall be measured using the calibrated reticle of the telescope. The telescope shall then be repositioned to focus directly on the target and the angular size of the target shall be measured. The magnification shall be calculated as the ratio of the first to the second angular measurements taken. The magnification shall be 1 $\pm$ 7%.

4.6.8 Entrance pupil. To determine conformance to 3.4.4, the integrated system configuration shall be used. With the G/VLLD pupil matched to the optical coupler, a uniformly illuminated target, subtending an angle larger than the field-of-view of the G/VLLD, shall be positioned in front of the G/VLLD. An optical magnifier shall be placed behind the eyepiece of the ITA so the exit pupil is brought into sharp focus in the plane of the reticle of the magnifier. Measure and record the diameter of the exit pupil by using

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the reticle in the magnifier. The G/VLLD shall be removed and the uniformly illuminated target shall be positioned in front of the optical coupler so it covers the field of view of the coupler. Measure and record the diameter of the ITA exit pupil. Reposition the illuminated target in front of the night sight objective lens of the ITA and switch the ITA to the night position. Again measure and record the diameter of the ITA exit pupil. The three exit pupil measurements shall not be less than 0.22 in (5.5 mm).

4.6.9 Field of view. To determine conformance to 3.4.5, the integrated system configuration shall be used. The target image shall be placed at one edge of the G/VLLD optical field and then at the opposite edge. The angular distance thrus traversed shall be measured. The field of view shall not be less than 5 degrees (0.087 radians) in any pupil matched condition.

4.6.10 Lateral color. To determine conformance to 3.4.6, the integrated system configuration shall be used. A monochromatic target [0.0000192 in or 0.0000258 in (486 nm or 656 nm)] shall be positioned in front of the G/VLLD and off-axis more than half the angular distance from the central optical axis to the edge of the field. A 5 nm diameter aperture shall then be positioned at the ITA exit pupil. An alignment telescope shall then be positioned behind the aperture, so the target image is located at the intersection of the telescope reticle crosshairs. The F and C filters [0.0000192 in and 0.0000258 in (486 nm and 656 nm)] shall be alternately positioned to provide alternate monochromatic illumination of the target. The apparent angular displacement of the target image which is caused by the alternate illumination shall be measured with the telescope. The lateral color of the integrated system shall be no greater than 3 minutes of an arc for wavelengths between 0.0000192 in and 0.0000258 in (486 nm and 656 mm).

4.6.11 Visual anomalies. To determine conformance to 3.4.7, the integrated system configuration shall be used. A 100 watt incandescent lamp shall be positioned at various angular locations outside the G/VLLD field of view but within the G/VLLD forward hemisphere at a distance of not more than 2 feet (610 mm). The ITA eyepiece apparent field shall be visually examined for evidence of flare or ghost images. The positioning of the lamp at various angular locations shall be repeated while the ITA eyepiece is observed for evidence of light emerging from outside the exit pupil. No visually perceptible defects shall be observed.

4.6.12 Temperature test. To determine conformance to 3.4.8.1, the optical coupler shall be tested as follows:

a. Operation

1. Low temperature. The optical coupler shall be tested in accordance with MIL-STD-810, Method 502.2, Procedure II at -25°F (-32°C).

2. High temperature. The optical coupler shall be tested in accordance with MIL-STD-810, Method 501.2, Procedure II at 145°F (63°C).

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b. Storage

1. Low temperature. The optical coupler shall be tested in accordance with MIL-STD-810, Method 502.2, Procedure I at 40°F (-40°C) for 6 hours.

2. High temperature. The optical coupler shall be tested in accordance with MIL-STD-810, Method 501.2, Procedure I at 160°F (71°C) for 4 hours.

4.6.13 Temperature shock test. To determine conformance to 3.4.8.2, the optical coupler shall be tested in accordance with MIL-STD-810, Method 503.2, Procedure I, except that the low temperature shall be -40°F (-40°C). To verify operational integrity, the resolution test (see 4.6.5), astigmatism test (see 4.6.6) and pressurization test (see 4.6.4) shall be performed after this test.

4.6.14 Humidity test. To determine conformance to 3.4.8.3, the optical coupler shall be operated during the humidity test in accordance with MIL-STD-810, Method 507.2, Procedure II, with the exception that the high temperature shall be 145°F (63°C) and the low temperature shall be -25°F (-32°C) and the number of test cycles shall be ten. The test shall be repeated with a low relative humidity of 5% at temperatures of +70°F (21°C) and +145°F (63°C).

4.6.15 Pressure test (altitude). To determine conformance to 3.4.8.4, the optical coupler shall be tested in accordance with MIL-STD-810, Method 500.2, as follows:

a. Procedure I. The air pressure shall be equivalent to an altitude of 40 000 ft (12.2 km).
b. Procedure II. The air pressure shall be equivalent to an altitude of 10 000 feet (3.1 km)

4.6.16 Rain test (nonoperating). To verify conformance to 3.4.8.5, a rain test shall be performed in accordance with MIL-STD-810, Method 506.2, Procedure I, with the exception that the test conditions shall be as specified in 3.4.8.5.

4.6.17 Sand and dust test. To determine conformance to 3.4.8.6, the optical coupler shall be operated during a sand and dust test in accordance with MIL-STD-810, Method 510.2, Procedure I, with the exception that sand and dust particle size shall be as specified in 3.4.8.6, with a density of 0.000000006 grams per cubic centimeter (gm/cm^3) and the temperature in test step 5 shall be 145°F (63°C).

4.6.18 Fungus test (nonoperating). To verify conformance to 3.4.8.7, the optical coupler shall be tested at 84°F and 95% RH with fungi as specified in MIL-STD-810, Method 508.3, for a period of 90 days. No test shall be required if the manufacturer can certify that only non-nutrient materials, as identified in MIL-STD-454, Requirement 4, have been used; or if the manufacturer can present certification of anti-fungal treatment acceptable to the Government.

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4.6.19 Vibration test (nonoperating). To verify conformance to 3.4.8.8, the optical coupler shall be tested to the requirements of MIL-STD-810, Method 514.3, Procedure I, Category 8, with the test level as specified in figure 3 with a sinusoidal cycling time of 3 hours per axis. Sweep time shall be 15 minutes from 5 to 500 to 5 Hz. Vibration amplitudes shall be as specified in 3.4.8.8.

4.6.20 Shock test (nonoperating). To verify conformance to 3.4.8.9, the optical coupler shall be tested to the requirements of MIL-STD-810, Method 516.3, Procedure I. The 11 ms sawtooth pulse with 40 g peak of figure 516.3-4 shall be used.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The optical coupler, 12310400, is intended for use on the M981 Fire Support Team Vehicle (FISTV) to interface between the G/VLLD and ITA.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Part number of the optical coupler required (see 3.3).
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If preproduction inspection is required (see 4.4.3).
h. If initial production inspection is not required (see 4.4.4).
i. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and acquisition of the first article (see 3.1).

6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.3).

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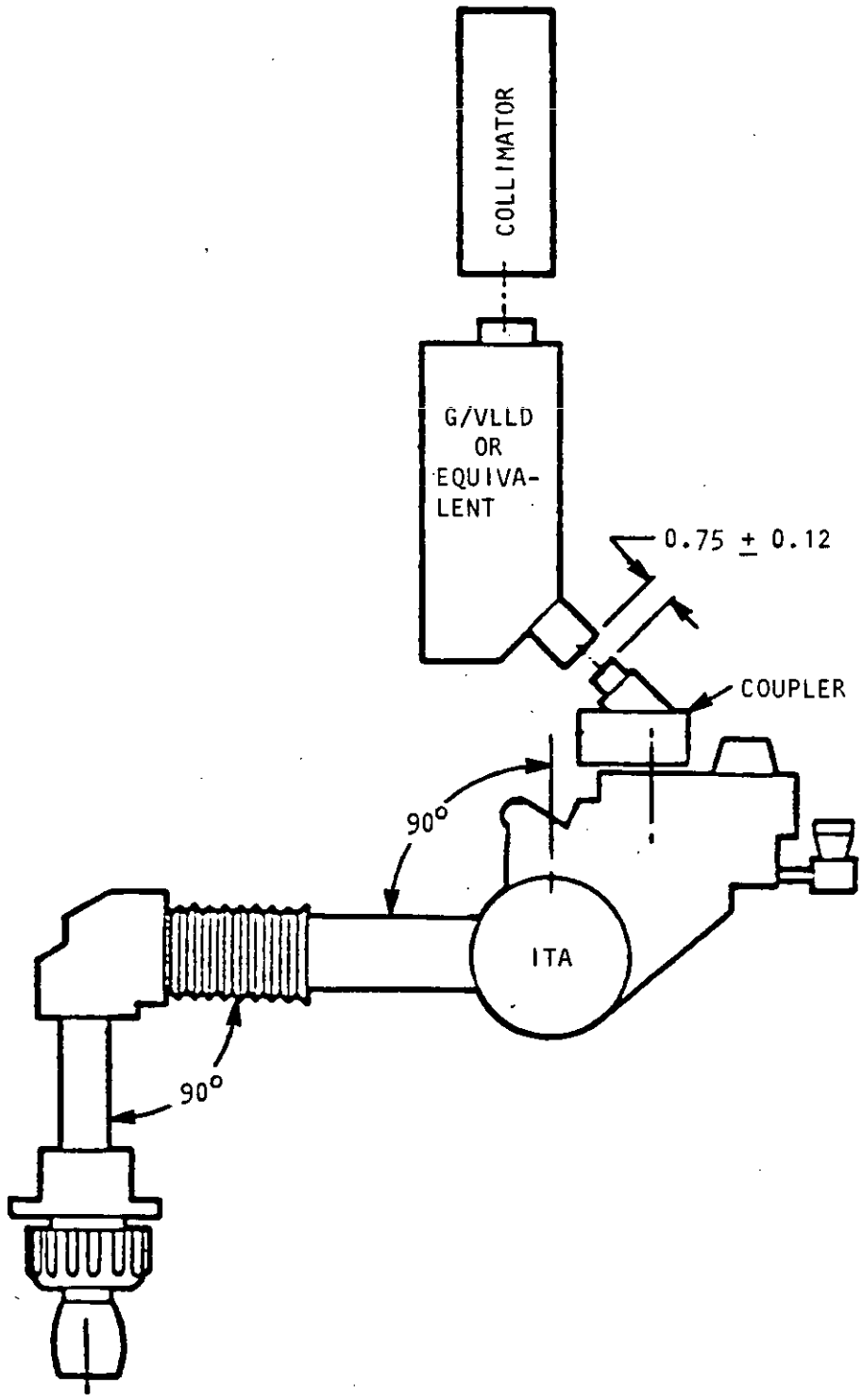
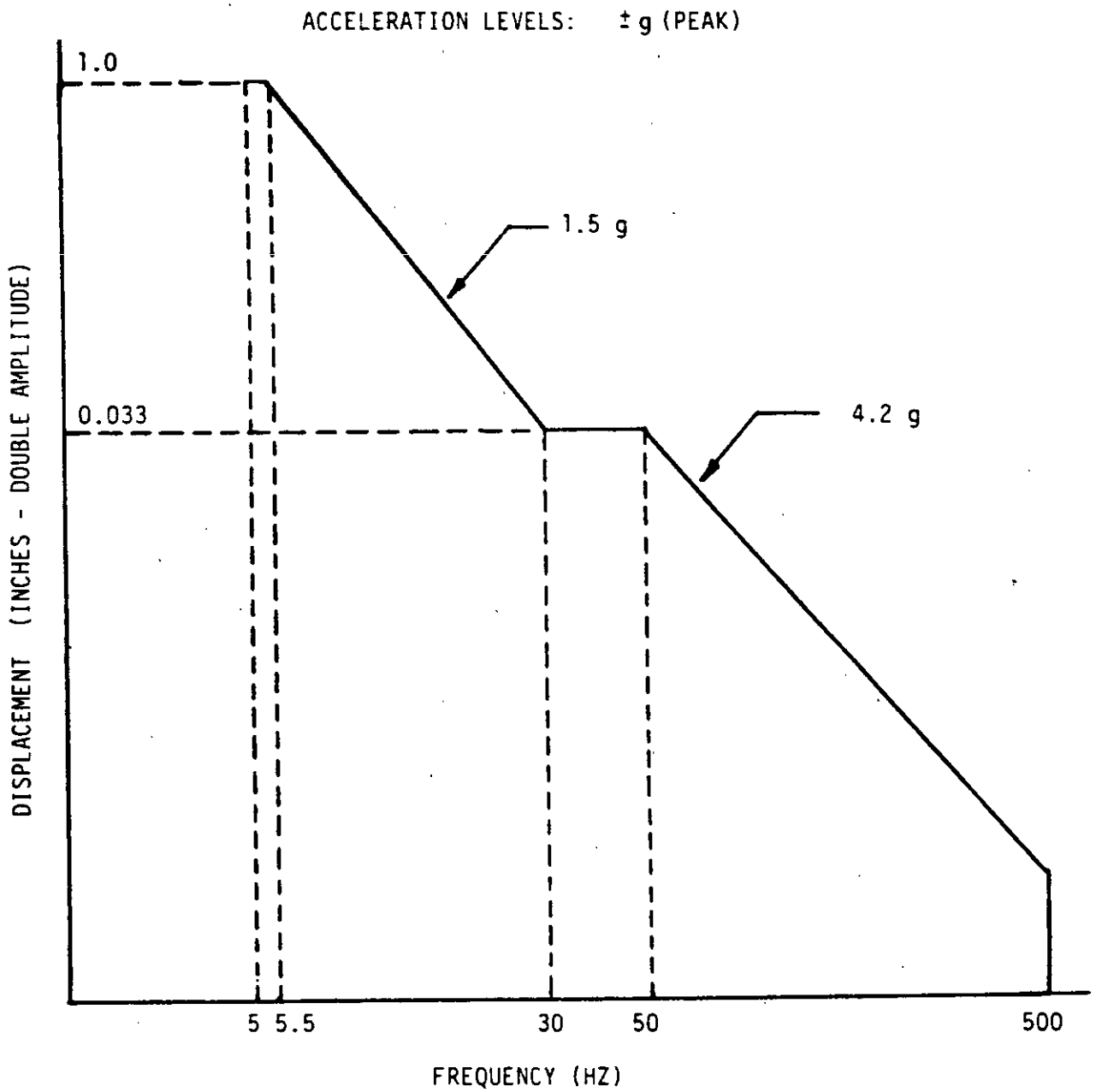


FIGURE 2. Integrated system test configuration.

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NOTE: All curves shall be extended to 2 Hz. when item resonances below 5 Hz. are expected.

FIGURE 3. Vibration test curve.

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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